

Compact Equations of Motion

Equation	Compact Form
Cabin Bounce (z_c)	$m_c \ddot{z}_c + \sum C_c(\dot{z}_c - \dot{z}_s) + \sum K_c(z_c - z_s)$ + Pitch/Roll Terms = 0
Cabin Pitch (θ_c)	$I_{yy} \ddot{\theta}_c + \text{Damping Coupling} + \text{Stiffness Coupling}$ - $M_c g h_{cp} \theta_c = 0$
Cabin Roll (ϕ_c)	$I_{xx} \ddot{\phi}_c + \text{Bounce/Pitch Coupling} + \text{Roll Terms} = 0$
Sprung Mass (z_s)	$M_s \ddot{z}_s + K_f(z_s - a\theta_s - z_{1f}) + C_f(\dot{z}_s - a\dot{\theta}_s - \dot{z}_{1f}) + \sum_{i=2}^3 (K_i + C_i) = 0$
Sprung Mass (θ_s)	$I_{syy} \ddot{\theta}_s + I_{syz} \ddot{\phi}_s - a(F_f) + \sum_{i=2}^3 l_i(F_i) = 0$
Sprung Mass (ϕ_s)	$-(I_{sxx} + m_s h_1^2) \ddot{\phi}_s + I_{sxy} \ddot{\theta}_s = -m_s g h_1 \phi_s$ + Roll Suspension Terms
Geometric Relations	$(z_s + l_i \theta + S_{tfi} \phi) - (L_D + \Delta l) \sin(\beta_i - \theta) = (z_{us})_i$

Table 1: Summary of Compact Equations of Motion

Final Set of Equations

1. Cabin Bounce (z_c)

$$\begin{aligned}
& m_c \ddot{z}_c + (C_{cfl} + C_{cfr} + C_{crl} + C_{crr}) \dot{z}_c \\
& - (C_{cfl} + C_{cfr} + C_{crl} + C_{crr}) \dot{z}_s \\
& + (K_{cfl} + K_{cfr} + K_{crl} + K_{crr}) z_c \\
& - (K_{cfl} + K_{cfr} + K_{crl} + K_{crr}) z_s \\
& - (C_{cfl} l_{cfeg} + C_{cfr} l_{cfeg} - C_{crl} l_{creg} + C_{crr} l_{creg}) \dot{\theta}_c \\
& - (-C_{cfl} l_{cf} + C_{cfr} l_{cf} - C_{crl} l_{cr} - C_{crr} l_{cr}) \dot{\theta}_s \\
& - (-C_{cfl} b - C_{cfr} a - C_{crl} b - C_{crr} a) \dot{\phi}_c \\
& - (C_{cfl} b - C_{cfr} a + C_{crl} b - C_{crr} a) \dot{\phi}_s \\
& - (K_{cfl} l_{cfeg} + K_{cfr} l_{cfeg} - K_{crl} l_{creg} - K_{crr} l_{creg}) \theta_c \\
& - (-K_{cfl} l_{cf} - K_{cfr} l_{cf} - K_{crl} l_{cr} - K_{crr} l_{cr}) \theta_s \\
& - (-K_{cfl} b + K_{cfr} a - K_{crl} b + K_{crr} a) \phi_c \\
& - (K_{cfl} b - K_{cfr} a + K_{crl} b - K_{crr} a) \phi_s = 0
\end{aligned} \tag{1}$$

Cabin Pitch (θ_c)

$$\begin{aligned}
& I_{yy_c} \ddot{\theta}_c - (C_{cfl} l_{cfeg} + C_{cfr} l_{cfeg} - C_{crl} l_{creg} - C_{crr} l_{creg}) \dot{z}_c \\
& - (-C_{cfl} l_{cfeg} - C_{cfr} l_{cfeg} - C_{crl} l_{creg} - C_{crr} l_{creg}) \dot{z}_s \\
& - (-C_{cfl} l_{cfeg}^2 - C_{cfr} l_{cfeg}^2 - C_{crl} l_{creg}^2 - C_{crr} l_{creg}^2) \dot{\theta}_c \\
& - (C_{cfl} l_{cfeg} l_{cf} + C_{cfr} l_{cfeg} l_{cf} - C_{crl} l_{creg} l_{cr} - C_{crr} l_{creg} l_{cr}) \dot{\theta}_s \\
& - (-C_{cfl} l_{cfeg} b + C_{cfr} l_{cfeg} a - C_{crl} l_{creg} b + C_{crr} l_{creg} a) \dot{\phi}_c \\
& - (C_{cfl} l_{cfeg} b - C_{cfr} l_{cfeg} a + C_{crl} l_{creg} b - C_{crr} l_{creg} a) \dot{\phi}_s \\
& - (K_{cfl} l_{cfeg} + K_{cfr} l_{cfeg} - K_{crl} l_{creg} - K_{crr} l_{creg}) z_c \\
& - (-K_{cfl} l_{cfeg} - K_{cfr} l_{cfeg} - K_{crl} l_{creg} - K_{crr} l_{creg}) z_s \\
& - (-K_{cfl} l_{cfeg}^2 - K_{cfr} l_{cfeg}^2 - K_{crl} l_{creg}^2 - K_{crr} l_{creg}^2 + M_c g h_{cp}) \theta_c \\
& - (K_{cfl} l_{cfeg} l_{cf} + K_{cfr} l_{cfeg} l_{cf} - K_{crl} l_{creg} l_{cr} - K_{crr} l_{creg} l_{cr}) \theta_s \\
& - (-K_{cfl} l_{cfeg} b + K_{cfr} l_{cfeg} a - K_{crl} l_{creg} b + K_{crr} l_{creg} a) \phi_c \\
& - (K_{cfl} l_{cfeg} b - K_{cfr} l_{cfeg} a + K_{crl} l_{creg} b - K_{crr} l_{creg} a) \phi_s = 0
\end{aligned} \tag{2}$$

3. Cabin Roll (ϕ_c)

$$\begin{aligned}
I_{xxc}\ddot{\phi}_c - (-C_{cfl}b + C_{cfr}a - C_{crl}b + C_{crr}a)\dot{z}_c \\
- (C_{cfl}b - C_{cfr}a + C_{crl}b - C_{crr}a)\dot{z}_s \\
- (-C_{cfl}l_{cfcg}b - C_{cfr}l_{cfcg}a + C_{crl}l_{crcg}b + C_{crr}l_{crcg}a)\dot{\theta}_c \\
- (C_{cfl}l_{cfcg}b + C_{cfr}l_{cfcg}a - C_{crl}l_{crcg}b - C_{crr}l_{crcg}a)\dot{\theta}_s \\
- (-C_{cfl}b^2 + C_{cfr}a^2 - C_{crl}b^2 + C_{crr}a^2)\dot{\phi}_c \\
- (C_{cfl}b^2 - C_{cfr}a^2 + C_{crl}b^2 - C_{crr}a^2)\dot{\phi}_s \\
- (-K_{cfl}b + K_{cfr}a - K_{crl}b + K_{crr}a)z_c \\
- (K_{cfl}b - K_{cfr}a + K_{crl}b - K_{crr}a)z_s \\
- (-K_{cfl}l_{cfcg}b - K_{cfr}l_{cfcg}a + K_{crl}l_{crcg}b + K_{crr}l_{crcg}a)\theta_c \\
- (K_{cfl}l_{cfcg}b + K_{cfr}l_{cfcg}a - K_{crl}l_{crcg}b - K_{crr}l_{crcg}a)\theta_s \\
- (-K_{cfl}b^2 + K_{cfr}a^2 - K_{crl}b^2 + K_{crr}a^2)\phi_c \\
- (K_{cfl}b^2 - K_{cfr}a^2 + K_{crl}b^2 - K_{crr}a^2)\phi_s = 0
\end{aligned} \tag{3}$$

4. Sprung Mass Bounce

$$\begin{aligned}
M_s\ddot{z}_s + K_f(z_s - a\theta_s - z_{1f}) + C_f(\dot{z}_s - a\dot{\theta}_s - \dot{z}_{1f}) \\
+ K_2(z_s - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta_s) \\
+ C_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}_s) \\
+ K_3(z_s - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta_s) \\
+ C_3(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}_s) = 0
\end{aligned} \tag{4}$$

5. Sprung Mass Pitch

$$\begin{aligned}
I_{syy}\ddot{\theta}_s + I_{syx}\ddot{\phi}_s - a[K_f(z_s - a\theta_s - z_{1f}) + C_f(\dot{z}_s - a\dot{\theta}_s - \dot{z}_{1f})] \\
+ l_2[K_2(z_s - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta_s) \\
+ C_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}_s)] \\
+ l_3[K_3(z_s - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta_s) \\
+ C_3(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}_s)] = 0
\end{aligned} \tag{5}$$

6. Sprung Mass Roll

$$\begin{aligned}
-(I_{sxx}\ddot{\phi}_s + m_sh_1^2\ddot{\phi}_s) + I_{sxy}\ddot{\theta}_s = -m_sgh_1\phi_s \\
+ k_{tf}(\phi_s - \phi_f) + c_{tf}(\dot{\phi}_s - \dot{\phi}_f) \\
+ k_{r1}(\phi_s - \phi_{r1} - \phi_{NRS}) + c_{r1}(\dot{\phi}_s - \dot{\phi}_{r1}) \\
+ k_{r2}(\phi_s - \phi_{r2} - \phi_{NRS}) + c_{r2}(\dot{\phi}_s - \dot{\phi}_{r2})
\end{aligned} \tag{6}$$

where, for $i = 2, 3$

$$\phi_{NRS} = (\beta_{Li}L_{DLi} - \beta_{Ri}L_{DRi})/S_{tfi} \tag{7}$$

Geometric Relations

$$(z_s + l_{i2}\theta + S_{tf2}\phi) - (L_D + \Delta l)\sin(\beta_{2L} - \theta) = (z_{us})_{2L} \tag{8}$$

$$(z_s + l_{i3}\theta + S_{tf3}\phi) - (L_D + \Delta l)\sin(\beta_{3L} - \theta) = (z_{us})_{3L} \tag{9}$$

4. Sprung Mass Bounce

$$\begin{aligned}
M_s \ddot{z}_s &- (C_{cfl} + C_{cfr} + C_{crl} + C_{crr}) \dot{z}_c \\
&- (-C_{cfl} l_{cfeg} - C_{cfr} l_{cfeg} + C_{crl} l_{creg} + C_{crr} l_{creg}) \dot{\theta}_c \\
&- (-C_{cfl} - C_{cfr} - C_{crl} - C_{crr}) \dot{z}_s \\
&- (C_{cfl} l_{cf} + C_{cfr} l_{cf} + C_{crl} l_{cr} + C_{crr} l_{cr}) \dot{\theta}_s \\
&- (K_{cfl} + K_{cfr} + K_{crl} + K_{crr}) z_c \\
&- (-K_{cfl} l_{cfeg} - K_{cfr} l_{cfeg} + K_{crl} l_{creg} + K_{crr} l_{creg}) \theta_c \\
&- (-K_{cfl} - K_{cfr} - K_{crl} - K_{crr}) z_s \\
&- (K_{cfl} l_{cf} + K_{cfr} l_{cf} + K_{crl} l_{cr} + K_{crr} l_{cr}) \theta_s \\
&+ K_f (z_s - l_f \theta_s - z_{1f}) + F_{df} \\
&+ K_2 (z_s - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR2} + l_2 \theta_s) \\
&+ C_2 (\dot{z}_s - \dot{z}_2 + l_2 \dot{\theta}_s) \\
&+ K_3 (z_s - z_3 - \beta_{L3} L_{DL3} - \beta_{R3} L_{DR3} + l_3 \theta_s) \\
&+ C_3 (\dot{z}_s - \dot{z}_3 + l_3 \dot{\theta}_s) = 0
\end{aligned} \tag{10}$$

5. Sprung Mass Pitch

$$\begin{aligned}
I_{syy} \ddot{\theta}_s + I_{syx} \ddot{\phi}_s &- (C_{cfl} l_{cfeg} + C_{cfr} l_{cfeg} - C_{crl} l_{creg} - C_{crr} l_{creg}) \dot{z}_c \\
&- (-C_{cfl} l_{cfeg}^2 - C_{cfr} l_{cfeg}^2 - C_{crl} l_{creg}^2 - C_{crr} l_{creg}^2) \dot{\theta}_c \\
&- (-C_{cfl} l_{cf} - C_{cfr} l_{cf} - C_{crl} l_{cr} - C_{crr} l_{cr}) \dot{z}_s \\
&- (C_{cfl} l_{cfeg} l_{cf} + C_{cfr} l_{cfeg} l_{cf} - C_{crl} l_{creg} l_{cr} - C_{crr} l_{creg} l_{cr}) \dot{\theta}_s \\
&- (K_{cfl} l_{cf} + K_{cfr} l_{cf} + K_{crl} l_{cr} + K_{crr} l_{cr}) z_c \\
&- (-K_{cfl} l_{cfeg} l_{cf} - K_{cfr} l_{cfeg} l_{cf} + K_{crl} l_{creg} l_{cr} + K_{crr} l_{creg} l_{cr}) \theta_c \\
&- (-K_{cfl} l_{cf} - K_{cfr} l_{cf} - K_{crl} l_{cr} - K_{crr} l_{cr}) z_s \\
&- (K_{cfl} l_{cf}^2 + K_{cfr} l_{cf}^2 + K_{crl} l_{cr}^2 + K_{crr} l_{cr}^2) \theta_s \\
&- l_f [K_f (z_s - l_f \theta_s - z_{1f}) + F_{df}] \\
&+ l_2 [K_2 (z_s - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR2} + l_2 \theta_s) \\
&\quad + C_2 (\dot{z}_s - \dot{z}_2 + l_2 \dot{\theta}_s)] \\
&+ l_3 [K_3 (z_s - z_3 - \beta_{L3} L_{DL3} - \beta_{R3} L_{DR3} + l_3 \theta_s) \\
&\quad + C_3 (\dot{z}_s - \dot{z}_3 + l_3 \dot{\theta}_s)] = 0
\end{aligned} \tag{11}$$